



1st Exercise

Architecture and CAD for FPGAs

Lookup table functions are given by their truth table, usually specified as an init value (provided as a hex or binary string). For this exercise, we assume a LUT-4 that has 16 configuration init bit values. All init strings shall be arranged MSB down to LSB (e.g., the LUT entry for all inputs=0 is the last bit in the init value string).

Aufgabe 1 (Lookup Table Don't-care Inputs)

a literal is a don't care (unused) if flipping that
bit does not change the output.

for a LUT4 (16 init values, we have to perform the
following compares:

# 0	0--+	0--+	0--+	0--+
#				
# 1	0- -+	0- -+	0- -+	1--+
#				
# 2	0- - -+	0- - -+	1-+	0--+
#				
# 3	0- - - -+	0- - - -+	1-+-	1--+
#				
# 4	0- - - - -+	1-+	0-+	0--+
#				
# 5	0- - - - - -+	1----+	0- -+	1--+
#				
# 6	0- - - - - - -+	1-----+	1-+	0--+
#				
# 7	0- - - - - - - -+	1-----+	1-+-	1--+
#				
# 8	1-+	0-+	0-+	0--+
#				
# 9	1----+	0- -+	0- -+	1--+
#				
# A	1-----+	0- - -+	1-+	0--+
#				
# B	1-----+	0- - - -+	1-+-	1--+
#				
# C	1-----+	1-+	0-+	0--+
#				
# D	1-----+	1---+	0- -+	1--+
#				
# E	1-----+	1-----+	1-+	0--+
#				
# F	1-----+	1-----+	1-+-	1--+

- a) Write a function/program that gets in an init string and that returns the dependent variables. For instance, if someone has implemented a 3-input Boolean function in our LUT-4, you should detect which LUT input is unused
- b) Try your program on different following input functions:
- $$f(a,b,c,d) = ((a \text{ AND } \bar{b}) \text{ OR } c) \text{ XOR } d$$
- $$f(a,b,c,d) = (((a \text{ AND } \bar{b}) \text{ OR } c) \text{ XOR } d) \text{ AND } (((\bar{a} \text{ AND } \bar{b}) \text{ OR } c) \text{ XOR } d)$$